

Local and Dyadic Interactions Enrich Evolutionary Dynamics in Structural Cellular Hash Chemistry



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Background: Hash Chemistry is a family of minimalistic artificial chemistry-based evolutionary models in which a deterministic hash function assigns a scalar score to entities of arbitrary size, opening a combinatorially vast possibility space (a “cardinality leap”) [1]. Since its introduction, the idea has been realized in several settings, from the original spatial formulation [1] to a fast non-spatial variant [2] and then, most recently, to Structural Cellular Hash Chemistry (SCHC) [3] that simulates evolution of self-replicating patterns on a $L \times L$ 2D cellular grid ($L = 100$ in this study).

Problem: In the original SCHC, the competition between two replicating structures was determined solely by individually evaluated hash values, and the structures were randomly selected for competition from anywhere in the space. Ecological compatibility between different patterns or their spatial proximity did not matter, and therefore, the competition always took place globally and individually. These assumptions grossly ignored the possible importance of spatial locality and dyadicity of ecological interactions.

Method: We extend SCHC by incorporating spatial locality and dyadicity of interactions among replicating patterns. In the extended SCHC, two adjustable parameters are introduced (Fig. 1): **spatial interaction range D** that limits the distance between two competing patterns, and **dyadic interaction probability P** that determines the likelihood for two competing patterns to be evaluated by the hash function *together* rather than independently. We conducted parameter-sweep numerical experiments to investigate the effects of D and P on SCHC’s evolutionary dynamics.

Results: Numerical simulations revealed that incorporating spatial locality and dyadicity significantly enriches SCHC’s evolutionary dynamics (Figs. 2, 3, 4). Specifically, spatial locality facilitates successful population growth (Fig. 2a) and spontaneous increase of self-replicating pattern size (Fig. 2b), while the possibility of dyadic interactions promotes the exploration of a greater number of replicating pattern types (Fig. 2c). We identified specific parameter settings for D and P at which locality and dyadicity lead to substantially enhanced successful population growth, increase of replicator size, and open-ended exploration compared to the original model setting (Fig. 3). These results demonstrate the importance of ecological interactions for open-ended evolutionary dynamics of artificial life systems.

Fig. 1: Schematic illustration of the extended SCHC with spatial and dyadic interactions.

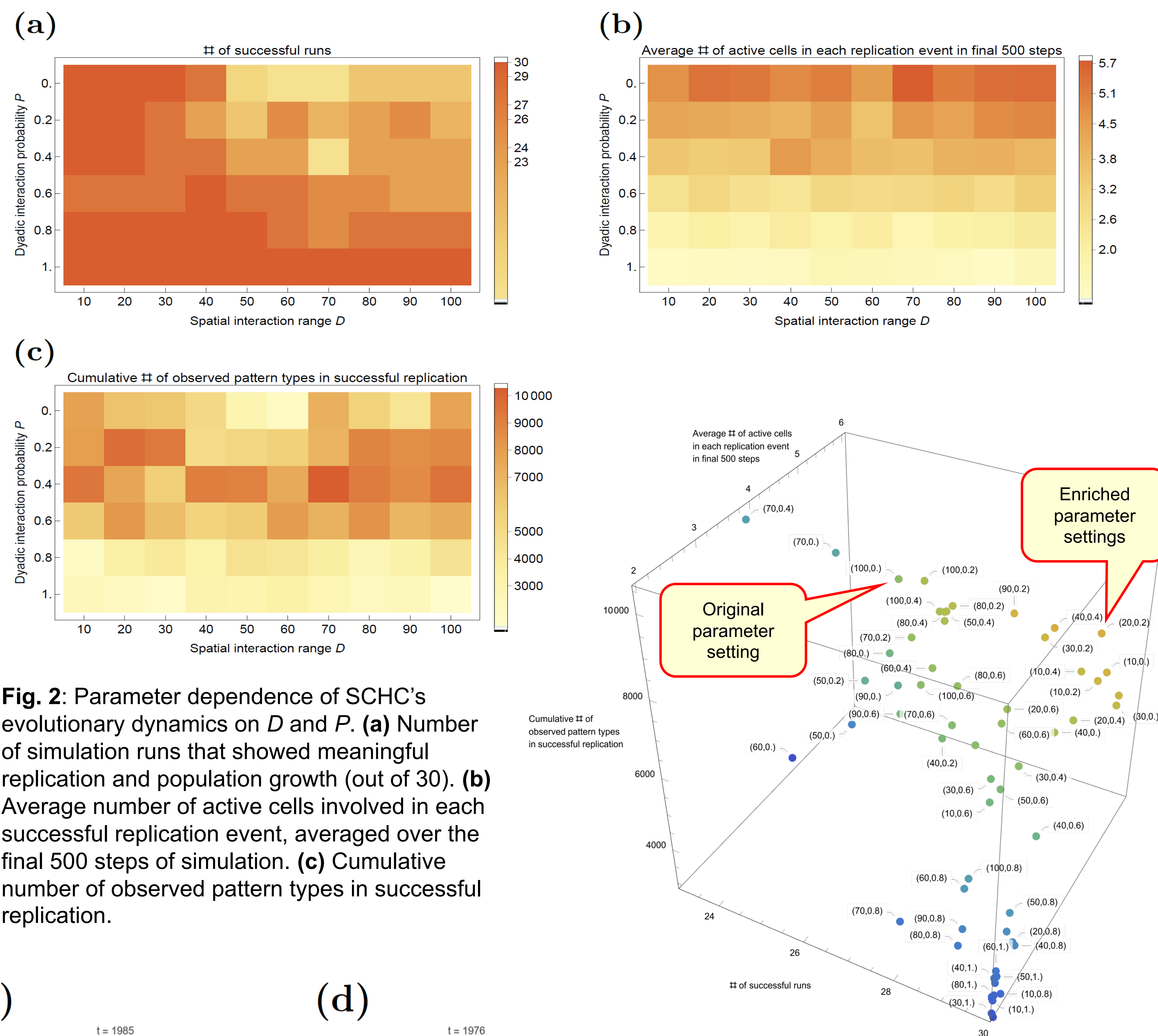
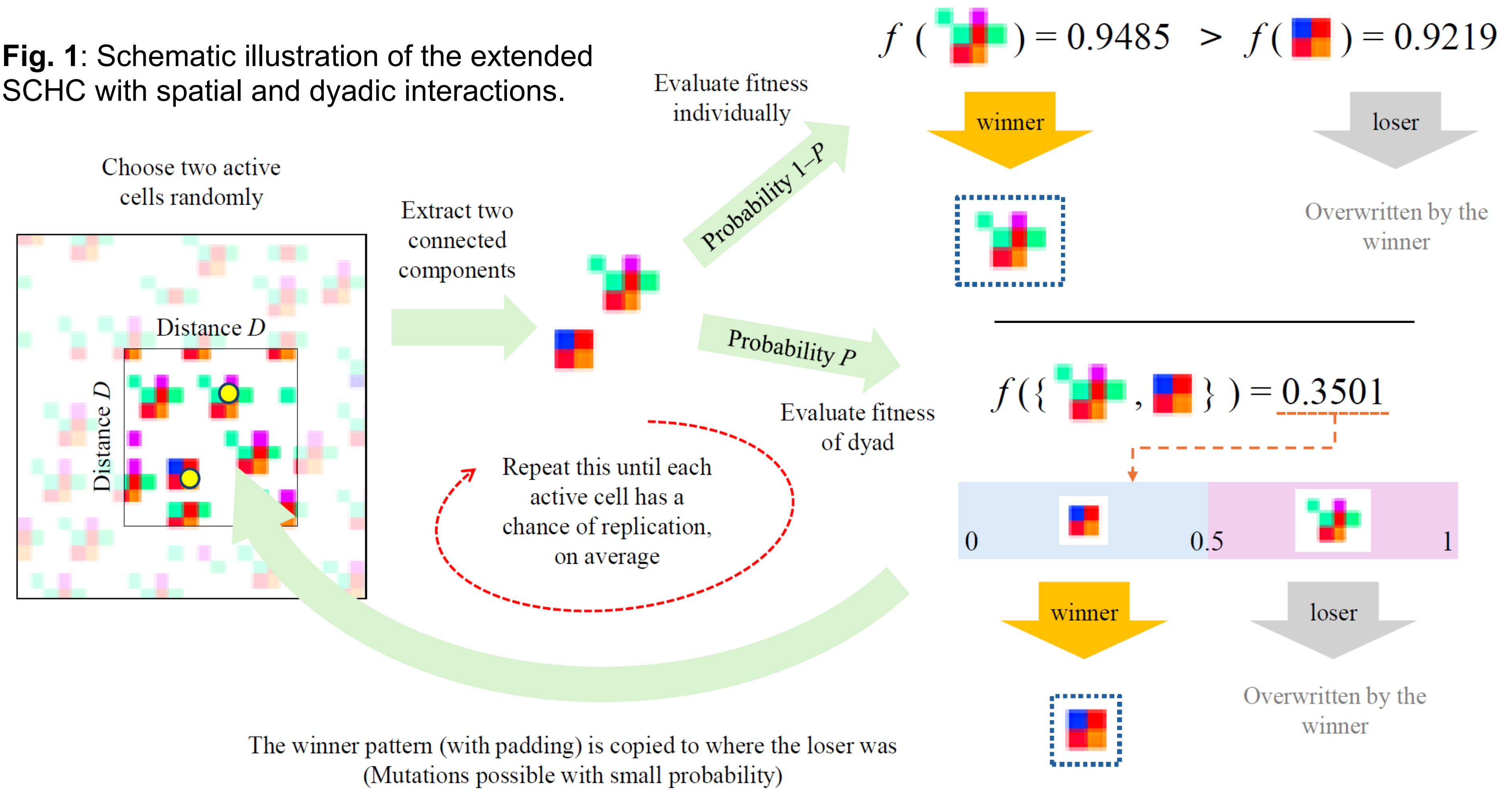


Fig. 2: Parameter dependence of SCHC’s evolutionary dynamics on D and P . (a) Number of simulation runs that showed meaningful replication and population growth (out of 30). (b) Average number of active cells involved in each successful replication event, averaged over the final 500 steps of simulation. (c) Cumulative number of observed pattern types in successful replication.

Fig. 3: 3D scatter plot summarizing the outcome measurements shown in Fig. 2. Markers are colored according to their proximity to the ideal outcome (top right corner of the box; yellow/warm = better, blue/cold = worse). Their (D, P) values are shown in callout labels.

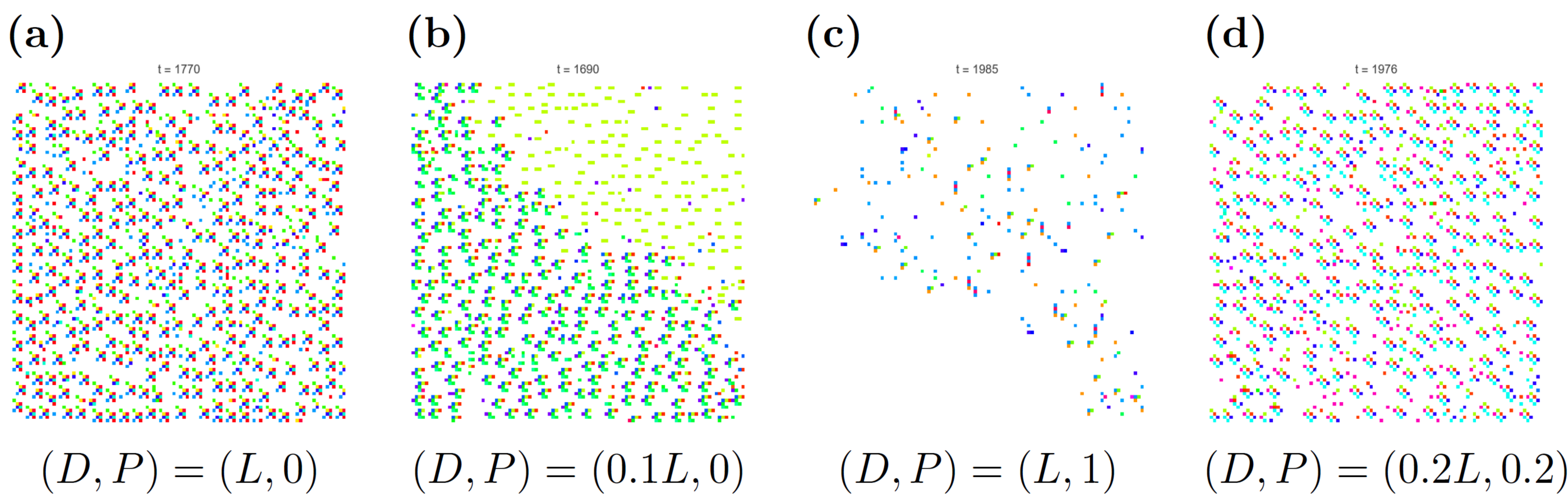


Fig. 4: Exemplar configurations of simulations under four selected parameter settings. (a) Interactions are always global and no dyadic interactions are used (original setting). (b) Interactions are limited only to spatial neighbors. (c) Interactions are always global and dyadic. (d) Interactions have both locality and dyadicity.

References

- [1] Sayama, H. (2019). Cardinality leap for open-ended evolution: Theoretical consideration and demonstration by Hash Chemistry. *Artificial Life*, 25(2), 104-116.
- [2] Sayama, H. (2024). Non-spatial hash chemistry as a minimalistic open-ended evolutionary system. In *Proc. 2024 IEEE CEC* (pp. 1-7). IEEE.
- [3] Sayama, H. (2025). Structural cellular hash chemistry. In *Proc. 2025 IEEE ALIFE-CIS* (pp. 1-7). IEEE.

Acknowledgments

This material is based upon work supported in part by the Waseda University Research Promotion Division Research Grant.

Videos



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